Scale Contrast Theory

Now as Relative-To

William P Robinson

October 12, 2026

Abstract

Every measurement is read as a ratio of marks: a distance, or a distance read against a distance, in the broad sense of §15. Nothing else enters. What measures and what is measured are one offer: distance contrasting distance.

Where the same ratio holds across the stations of one engagement, that ratio is its scale, and the scale is offered by the orientation of the reading, not found.

The now is a snapshot of c. A separation is read against the distance light spans in the interval, $\rho = \Delta x/(c\Delta t)$. Temporal scale is distance scaled: time is not a dimension beside distance, it is the canvas a separation is read against. The label of time sees time contrasted: the contrast of two times is a time, and no instant stands alone. Two orientations read one interval, and that interval is the one negotiation.

Where $\rho \leq 1$ the separation is within reach. Causality, if the word is wanted, is the now's: a rate held across seats and continued, not a cause sent from an earlier mark to a later one. Where $\rho > 1$ the separation is beyond reach, and the order of its two ends is offered by the orientation alone. The ratio is the same for either sign of Δt. Orientation offers the sign. The grammar does not derive the arrow.

The paper takes the form of a turn, a mark as a fraction of a whole, and sets the whole by light. Rooms run from the Sun's light interval to a capacitor's decay, with the system's own year as a tally. A Lorentz seating is named and not yielded from ρ alone. Every claim carries a label and the mark that would separate it. The reading is offered. The law is declined.

Reference

The first paper is *Scale Contrast Theory: A Grammar of Observation* (September 30, 2026). This paper is its sequel seating, with the same primitive: contrast is read at a now, and the now is a relative-to. The first paper's light-tied cut, $(v/c)^2(r/L) = 1/2$, is another room and is not used here. Shared letter c. No shared derivation.

Cohesion

The turn supplies the form: a mark is a fraction of a whole, ρ = mark/whole, and n steps of 360/n close the turn for every n. The now is that form with the whole set by light: the span, $c\Delta t$. Every claim below reads one whole at a seat. The seat changes the label and the number of steps, not the

whole and not the relation. Different marks stay different: 60 is 1/6 of the whole, and 360 is the whole. The gap between a discrete seat and the continuous turn differs by seat.

Two things hold throughout. ρ is invariant under a change of seat. The period of closure is relative to the engagement: 360 for direction, 720 for phase, none for a boost.

How to read

Offered means present to be read. *Received* means the reading of it. *Regard* is the rate returned. *Seat* means the canvas and the mark are named before the ratio is read. A *label after* is a name the reading does not need. *The law declined* means no equation is seated as the source.

Facing means ρ for one separation, $\rho = \Delta x/(c\Delta t)$, read in magnitude. *Reach* means $\rho \le 1$, and *beyond reach* means $\rho > 1$. *Now* means the slice an orientation reads as $\Delta t = 0$. It is a relative-to, not a moment on a line. *Temporal scale* means distance scaled. The *span* is the distance light spans in an interval, named $c\Delta t$ after the reading. A *wake* is a later reading taken as carried by an earlier one. A *factored wake* is that carrying written with one common rate, $(\Delta x_2 - \Delta x_1)/(c\rho)$, and it holds only while the rate is one rate.

Labels. *Shown* means a number or identity that can be checked. *Defined* means a convention. *Stance* means a position with no failing mark yet. *Bounded* means a claim whose counter-case is named. Each section ends with the mark that would separate it, or says that none can.

Notation. ρ without a subscript is the facing of a separation. ρ_t and ρ_σ are the marks of the close in §9. K is the tally v^2r of a station, in the system's own units. L is the canvas of a room: a length in the first paper's cut, a time in the close, and a span of 499 s for the Sun seen. The v of a station's traversal is not the v of a relative regard between orientations.

1. The now as snapshot of c

Isolation. This separation, read at a seat. Canvas: the span, the distance light spans in the interval. Mark: the separation Δx.

Form. $\rho = \Delta x/(c\Delta t)$. Reach is $\rho \le 1$. Temporal scale is not beside the separation. It is the canvas the separation is read against. Earlier and later are the same facing read twice.

Identity. Substitute the form: $\Delta x/(c\rho)$ returns Δt. The difference of two such readings returns the interval between the snapshots. It is the span already named, not a wake discovered. The factored form $(\Delta x_2 - \Delta x_1)/(c\rho)$ holds only while ρ is the same rate at both snapshots. Take 1.0 m of path through air and then 1.0 m through water, n = 1.333. The times are 3.34 ns and 4.45 ns, a total of 7.78 ns. The factored form with one rate, c, gives 2.0 m/c = 6.67 ns. The factored wake fails by 17%. The first paper's angle rows already seat that change of rate.

Standing. Since 1983 the metre has been defined as the distance light spans in 1/299,792,458 s. In that definition a distance is a time read through c. This paper claims no more than that definition already does. Its permission is narrower: no temporal dimension is seated beside distance, and c is a ratio read from the marks, not a constant prior to them. The ct convention of the standard account is the same bookkeeping.

Limit. ρ is a magnitude, so it is the same for Δt and −Δt. It gives reach, not order. No arrow is read from it.

Mark. A measured ρ different from 1 with a light signal, or a measured total through two media that equals the factored value.

2. One whole, many seats

Shown. ρ = mark/whole, and the parts sum to the whole. 360, 2π, 400, and 1 are seats for one whole, and ρ is invariant. 60 and 360 are not two wholes: 60 is one step of the whole, and 360 is the whole. At every seat n steps close it.

Seat	Step	ρ per step	Perimeter ÷ diameter	Unit-free gap
Triangle	120°	1/3	2.598	0.173
Square	90°	1/4	2.828	0.100
Hexagon	60°	1/6	3.000	0.045
Decagon	36°	1/10	3.090	0.016
Dodecagon	30°	1/12	3.106	0.011
360-gon	1°	1/360	3.14155	0.000013

The circle and its discrete readings share no common unit, since π is irrational, so every inscribed n-gon leaves a gap. The unit-free gap is $1 - n \sin(\pi/n)/\pi$, the shortfall of the perimeter as a fraction of the whole. At $n = 6$ it is $1 - 3/\pi = 0.045$, which is $\pi - 3 = 0.1416$ in diameter units. It is the hexagon's gap, not the circle's: the decagon's is 0.016. It is arithmetic and cannot drift. Writing the whole as 1440 units gives a hexagon step of 240 and a decagon step of 144, with ρ unchanged.

Limit. As n grows the gap falls as about $\pi^2/(6n^2)$, and as $n \to \infty$ it is zero. The circle is the limit that no finite seat reaches. It is the unread seat, and every polygon in the table is a reading of it.

Mark. An inscribed n-gon whose perimeter reaches πd. None: $n \sin(\pi/n)$ is below π for every n.

3. Unit seating

Shown. 60° is where the chord equals the radius, so six close the hexagon, and its perimeter is 3 diameters. 60 is not the smallest turn, since any divisor of 360 closes. What is special is that the chord, $2r \sin(\theta/2)$, reaches the radius only from 60° upward. At 36° the chord is 0.618 of the radius. So 60 is the smallest central angle whose chord is not shorter than the canvas's radius.

Mark. A central angle below 60° whose chord reaches the radius. None.

4. Why the whole is written 360

Shown. The whole turn is not exactly 360. It is written 360, and any N is a seat. What 360 offers is divisibility: it has 24 divisors, more than any smaller number, so a hexagon step (60), a decagon step (36) and many others are whole numbers. $361 = 19^2$ has three divisors, so its steps are whole only

for 19 equal parts. 400 gradians gives a hexagon step of 66.67. The ρ per step is the same in every case. Only the label changes.

Mark. A whole number below 360 with 24 or more divisors. None.

5. The sphere holds the family

Shown. A surface circle at angular radius α has C/d = π sin α/α. At α = π/n it equals the inscribed n-gon's perimeter over diameter, n sin(π/n). The hexagon sits at 30°, where C/d = 3.0000, and the square at 45°, where it is 2.8284. A slice seats one member. The sphere holds the family of discrete readings, each at its own α. A plane slice keeps C/d = π by its chord diameter, so the family is the surface reading. The identity ties the hexagon, the circle and the sphere together.

Mark. A surface circle at α = π/n with C/d other than n sin(π/n). None: it is an identity.

6. Seatings of time as distance

The Sun, seen. Canvas: the span. Marks: the distance $\Delta x = 1.496 \times 10^{11}$ m and the interval Δt. $\Delta t = \Delta x/c = 499$ s, 8 min 19 s. The Sun seen is the Sun at an offset of 499 s. Looking is reading a distance as an offset in time. This is L written as a span, the same office as the first paper.

A lunar echo. Marks: the Moon's mean distance 3.844×10^{8} m, and the round trip, twice that. $\Delta t = 2\Delta x/c \approx 2.56$ s. The interval measured and the distance returned are one ratio, not two quantities.

Earth's own traverse over the Sun's light interval. Canvas: the same 499 s. Mark: the distance Earth covers in that interval at 29.785 km/s, 1.486×10^{7} m. $\rho = v/c \approx 9.9 \times 10^{-5}$, well inside reach. It is measured: for v □ c, ρ in radians is the aberration of starlight, about 20.5″ (Bradley, 1729).

Room	Distance read (m)	Δt read	ρ
The Sun, seen	1.496×10^{11}	499 s	1.000
Earth's traverse, same interval	1.486×10^{7}	499 s	9.9×10^{-5}
Lunar echo, round trip	7.688×10^{8}	2.56 s	1.000

Seating. The light-tied rows are read with a light signal, so ρ = 1 is seated, not found. Earth's traverse is read from a measured angle, so its ρ is found, not seated. The rows show what a time is when read as distance. They do not show that light had to be the signal.

Mark. A measured aberration different from v/c.

7. The year as the tally

Shown. The Sun's room already offers a period: one full turn at 1 AU is one year. Take the year as the unit of time and the astronomical unit as the unit of distance. The tally the room returns is $K = v^2 r = (2\pi)^2 \cdot 1 = 4\pi^2$. It is not fetched. It is what the seating returns when the canvas is the year the system already is and the mark is the distance that year traverses. The current $GM = 4\pi^2$

AU^3/yr^2 is a later name for the same tally.

The first paper's scale is $L = 2K/c^2$. With $c = 63{,}241$ AU/yr, read from the 499 s span, $L = 8\pi^2/c^2 = 1.97 \times 10^{-8}$ AU = 2.95 km. One regard, two writings. No mass, no G. The numeral changes with the units and the relation does not.

Other stations keep the same tally and return different years. Mars, at 1.5237 AU and 1.8808 yr, gives $a^3/P^2 = 1.000$ in those units.

Mark. A station whose a^3/P^2 departs from 1 by more than the planetary perturbations. Earth's own $4\pi^2$ is fixed by the choice of units, so it is not a mark.

8. Full depends on engagement

Shown, measured. Direction closes at 360°. Spinor phase closes at 720°, with overlap $\cos(\theta/2)$. A boost never closes, since rapidity runs without bound. Three closures for three engagements, so full is engagement-relative. The half-speed wave, $\cos(\theta/2)$, has period 720. It is the same fact as the spinor phase, seen as a wave. Changing a wave's frequency or amplitude is a scale step, not a new relative.

The 720 is measured: neutron interferometry has confirmed the 720° spinor phase (Rauch et al., 1975; Werner et al., 1975). It is one relation, written two ways.

Mark. A neutron interferometer that finds a 360° period.

9. The interval, not the origin

Isolation. This capacitor through this resistor. Contrast: charge separation. Canvas: the initial charge q_0, and L, the room's own time, dual-named after the reading as RC. Marks: charge q, and the seat t. Voltage is a label after, q/C, the same ratio if C is not changing.

Form. $\rho_\sigma = q/q_0$ and $\rho_t = t/L$. The marks show $\rho_\sigma = e^{-\rho_t}$. They also show the interval property: $\rho_\sigma(t_1 + t_2) = \rho_\sigma(t_1) \cdot \rho_\sigma(t_2)$. The thinning over an interval does not depend on where the interval began. If ρ_σ is continuous, not zero, and depends only on the interval, it takes the form $e^{k\rho_t}$. The room seats $k < 0$ for the orientation that reads t forward from the charged seat, since the charge returns toward zero and does not grow, and L is the canvas that makes $k = -1$. Read from the other end, the same rows give $k > 0$. This is seated as a mark, not derived as a law.

Check. Read the envelope at t and at 2t. At equal intervals $\rho_\sigma(2t) = \rho_\sigma(t) \cdot \rho_\sigma(t)$. At the cut, $0.500 \times 0.500 = 0.250$. Where a measured product misses, the close is not the room's.

The label. Time as a scale of distance sees distance contrasted, offering a distance-scale. The label of time sees time contrasted. Seat the rate as canvas: the name offered to a distance is $\tau(\Delta x) = \Delta x/c$, or in this room $\rho_t = t/L$. Contrasted with itself, $\tau(\Delta x_2) - \tau(\Delta x_1) = \tau(\Delta x_2 - \Delta x_1)$, and likewise for ρ_t. The contrast of two times is a time, and no instant stands alone. This holds by seating, since the label is linear in the distance, and it holds within one orientation. Across orientations the label changes (§10). Addition of the label is multiplication of the close: $e^{-(\rho_1+\rho_2)} = e^{-\rho_1} \cdot e^{-\rho_2}$.

Seat	ρ_t	ρ_σ	Reading
0	0.000	1.000	the seat, unread as a start
0.288 L	0.288	0.750	one facing
0.693 L	0.693	0.500	the cut, charge half
0.693 L + 0.693 L	1.386	0.250	0.500 × 0.500

The rows are chosen and cannot fail the relation used to choose them. A measured engagement would replace them.

Standing. 1/L is what the room already is. Current, if named, is a label after the rate. The sign constituted on the interval is the route. The equation does not choose the sign. Orientation does. The room's dissipation, why this run and not its reverse, is a label after and is not explained here.

Limit. A second engagement is counted when a second canvas is named before the reading: a driven source, a second time constant, a dielectric with its own relaxation. The count is fixed by the seating, not by the misfit.

Mark. A measured $\rho_\sigma(2t)$ that is not $\rho_\sigma(t)^2$.

10. One interval, two readings

Isolation. Two events, read by one orientation as simultaneous ($\Delta t = 0$), separated by $\Delta x = 3.0 \times 10^8$ m. Canvas: the span, for any interval seated.

Form. For this orientation ρ is unbounded: the separation is beyond reach. A second orientation in uniform motion at v along the separation reads the same interval differently. The readings differ and the interval does not: $c^2\Delta t^2 - \Delta x^2$ has magnitude $(3.0 \times 10^8 \text{ m})^2$ in both.

v/c	γ	Δt′ (s)	Δx′ (m)
0.1	1.005	−0.10	3.02×10^8
0.5	1.155	−0.58	3.46×10^8
0.9	2.294	−2.07	6.88×10^8

Seating. The table checks the arithmetic of the seated Lorentz form, $\Delta t' = -\gamma v \Delta x/c^2$. It does not derive the form from ρ alone. Under reciprocity, homogeneity, isotropy and continuity the form follows from $\rho = 1$ for light held across orientations in uniform motion, and c is read from the marks. This v is the relative regard between orientations, not the traversal of a station. For $\Delta x > 0$ the sign of $\Delta t'$ is opposite the sign of v.

Two orientations read one interval. The interval is the one negotiation, and the Lorentz form is how the two readings relate.

Standing. For $\rho \le 1$ every orientation in uniform motion, on this seating, reads the same order. For $\rho > 1$ the order is offered by the orientation, and another orientation offers the other. Inside reach the order holds across orientations, and holding is what the first paper calls a scale constituted. Beyond reach nothing holds, and the order is that orientation's offer. This is why the now is a

relative-to: the snapshot of c is shared only where reach is.

Limit. Reach is a necessary condition. It does not say what within reach causes what. The seating is tested at one place: ρ = 1 for light holding across orientations. A measured dependence of c on an orientation's motion or direction would fail it.

Mark. An interval that differs between orientations.

11. The spiral is one offer

Geometry shown; constitution bounded. Seen end-on, the spiral's windings coincide as one facing. Seen side-on, they are distinct counts, and a helix side-on is a sine wave. 180 is the opposite facing. The facing repeats and the count is carried. The six stations, 180 to 1800, are listed by size, not in order. The constitution reading is bounded: one offer holds for every spiral generated by a single advancing axis. A spiral with a precessing or composite axis is the counter-case.

Mark. A spiral not generated by a single advancing axis. It exists, so the claim is bounded.

12. Scale step

Defined. The scale step changes the label and not ρ, and the radix is the seat. The step must scale the whole as well as the mark. In base ten it is ×10, and in degrees and minutes it is ×60. So 36(0) = 360 reads as ten steps of 36° closing the turn (the decagon in §2), not as an appended digit.

Mark. A scale step that changes ρ. None.

13. Big/small and close/far

Shown. Big/small and close/far are ratios against a named canvas. Far is read here as a ratio's distance from 1 in decades, $|\log_{10} \rho|$, which is a seat.

Canvas	Atom (10^{-10} m)	Moon (3.844×10^{8} m)	Farther
1 m	10.0 decades	8.6 decades	the atom
0.2 m	9.3 decades	9.3 decades	equal
0.1 m (a cup)	9.0 decades	9.6 decades	the Moon

The crossover is the geometric mean of the two sizes, $\sqrt{(10^{-10} \times 3.844 \times 10^{8})} \approx 0.2$ m. Above that canvas the atom is farther in scale than the Moon.

Mark. A crossover canvas not near 0.2 m.

14. The threshold and the spectrum

Stance. A ratio is one facing and a spectrum is many. When the ratio is lost to a spectrum, the bridge no longer exists. Physics names a version of this as scale separation: the description at one scale does not need the next. At a human canvas an atom is ten decades from 1 (§13). It exists, but it is

not within that spectrum. The grammar reads within a spectrum, and beyond it nothing is read. That is not the same as nothing being there.

Mark. None yet. A mark would be a ratio read across many decades with no change of seat.

Note. An earlier draft labeled this section Shown. It is a Stance: the threshold is a position with no failing mark yet, so Shown was not earned.

15. Distance

Defined. Distance is an expression sourced from a relation, relative to its seat and its orientation, with the metric read from marks. An interval is defined as a mark read against a named canvas, so no interval can be seated without one. That makes this a definition, not a finding. The open question is empirical: whether a seating exists in which a number is assigned with no canvas named. A dimensionless number is itself a ratio.

16. Continuations

Stance, with marks. Where a rate has been read from marks, it can be held across seats and continued. The continuation is the prediction: the now's rate written at a further seat, not a push from an earlier mark. The holding is the seat. The continuation assumes the rate is one rate, and it stops where a second rate enters.

Freefall. Canvas: the interval. Rate: the room's rate, read from freefall, 9.81 m/s². Mark: the distance fallen s. $s = \frac{1}{2}gt^2$, and g is a label after the reading. With H = 19.62 m the fall takes 2.00 s.

t (s)	s (m)	s/H	Reading
0.00	0.00	0.00	the seat
1.00	4.905	0.25	quarter
1.414	9.81	0.50	half
2.00	19.62	1.00	the mark

Capacitor, continued. The seat at 0.693 L offers $\rho_\sigma = 0.500$. The same rate held over another 0.693 L returns $0.500 \times 0.500 = 0.250$. The second seat is the first rate read over the added interval.

Light interval, continued. The 499 s is the now for 1 AU. Double the distance and the same rate returns 998 s.

Mark. A measured s that is not $\frac{1}{2}gt^2$ in vacuum. A second rate, drag, ends the continuation. That is the limit, not a failure of the seating.

17. What is not written

No Lagrangian. No action integrated as a source. No force yielded. No time as a dimension beside distance. The Lorentz form is not derived from ρ alone. Particles and fields are not seated. No mass is isolated, and no G is seated as a prior. An equation that produces the physics would be the

allowance this paper does not grant. Awareness is not derived and is not asked of the arithmetic.

What this path buys, and what it costs

This paper offers a path, not a destination. The legacy path reads a later mark as produced by an earlier one. This path reads each mark as a facing at a now, with a rate held across seats and continued. The rows are the same under both, so the paper does not claim that this path is true and the other false. It claims that the path is open.

It buys three things. No mass, force or time dimension is seated as a prior. The now is a relative-to, which matches the relativity of simultaneity: beyond reach there is no universal before and after, and the order that holds inside reach is the part of cause that the structure itself fixes. And every claim carries a mark that could separate it.

It costs three things. The arrow is not derived: orientation offers the sign, and the room's dissipation is a label after. The held rate does the work a law does, so the continuation of §16 is a Stance, and it stops where a second rate enters. And the path makes no prediction that the legacy path does not.

The path has precedent. Russell argued in 1913 that physics has functional relations and no use for cause (*On the Notion of Cause*), and Mach's reading of mechanics as relations between measured quantities is near it. What is offered here beyond them is a worked grammar: a seat named before every ratio, and a mark after every claim.

Mark. A reading where this path and the legacy path differ. None has been seated.

Readings

Reading	What is seated or read	What would separate it
Sun, $\rho = 1$	L as a span, 499 s	A measured ρ different from 1 with a light signal
Lunar echo, $\rho = 1$	Round trip, 2.56 s	A round-trip interval not equal to $2\Delta x/c$
Earth's traverse, $\rho \approx 10^{-4}$	A measured angle, $\rho = v/c$	A measured aberration different from v/c
Year as tally	$K = 4\pi^2$ in the system's units	A station whose a^3/P^2 departs from 1 beyond perturbations (Mars: 1.000)
Factored wake	1 m of air and 1 m of water: 7.78 ns, not 6.67 ns	A measured total equal to the factored value
Close at t and 2t	$0.500 \times 0.500 = 0.250$	A measured $\rho_\sigma(2t)$ that is not $\rho_\sigma(t)^2$
Freefall	$s = \frac{1}{2}gt^2$, continued from the now	A measured s that is not $\frac{1}{2}gt^2$ in vacuum
Spinor phase	720, one relation	A neutron interferometer that finds a 360° period
Interval	One interval, two readings	An interval that differs between orientations

The light-tied rows and the close are seatings. Earth's traverse and the spinor phase are measured. The year is the system's own period. The factored wake is a worked limit. Freefall is a continuation. None of them returns a wake the marks did not already contain.

Anchors

Anchor 1 is the seat table (§2). It is arithmetic: one whole, one closure relation, and a unit-free gap that shrinks as the seat is refined. It cannot drift. It is the floor.

Anchor 2 is the spinor 720 (§8). It is measured, and the ceiling is single: one relation, $\cos(\theta/2)$, written two ways, the phase and the wave. Its strength is the linkage, since that one relation fixes both the number 720 and the wave's period. It can drift only if the $\cos(\theta/2)$ relation fails. In that case the cohesion is restated without it and the arithmetic floor remains. The document is reduced, not separated.

The cohesion is not the two anchors as such. It is that ρ is invariant under a change of seat, that the period of closure depends on the engagement, and that the sphere identity (§5) ties the hexagon, the circle and the sphere together.

Standing

The now is a relative-to. It is the snapshot of c: a separation against the distance light spans. Temporal scale is distance scaled. The Sun seen is 499 s of distance, $\rho = 1$, and Earth's own traverse over that interval reads $\rho \approx 10^{-4}$, which the aberration of starlight measures. In the system's own units the year offers the tally $K = 4\pi^2$, and the scale is $L = 8\pi^2/c^2$. No mass, no G.

Within reach the order holds across orientations, on a seating named as such. Beyond reach it is offered by the orientation. The interval is one, and the readings differ. The capacitor shows the interval alone carries the thinning, checked by $\rho_\sigma(2t) = \rho_\sigma(t)^2$. A route may be constituted by a sign on that interval. It is not found there. Orientation offers the sign. The grammar does not derive the arrow.

Where a rate is held across seats, it is continued, and the continuation is the prediction. Where the rate changes, the factored wake does not hold. The grammar reads within a spectrum, and beyond it nothing is read.

The reading is offered. The route may be constituted. The law is declined.